FakeSpotter Arena

A game where students practice telling real posts from AI fakes, learn why they got fooled, and a teacher dashboard that shows who needs help. This brief explains the problem, every component the team should build, how to build each one in 6.5 hours, and how to pitch it.

Theme
TruthEngine for EdTech: multi-modal cognitive defense

Time
6.5 hours of building, then a short live pitch

Users
School and university students, their teachers. The jury plays it live.

1 · THE PROBLEM

The problem in plain words

Kids meet fakes online first

AI photos, edited images, AI-written posts and cloned voices reach students every day on social media and in group chats. School rarely covers it.

Lectures don't survive a good fake

"Look for weird hands" tips go stale. Modern fakes often have no visible flaws, so a slide deck doesn't help when a convincing one arrives.

Teachers can't see who's at risk

A lecture produces no data. The teacher doesn't know who believes fakes, who distrusts everything, or whether anyone improves.

Why this is hard

CHARACTERISTIC	WHAT IT MEANS FOR THE DESIGN
People can't see good fakes	Humans are near chance on high-quality AI images and voices. Teach **habits** (check the source, look for other reports), not only visual tricks.
Generators change every few months	Visual tells disappear. A fixed list of tips goes stale; verification habits don't.
Detectors are unreliable too	They help, but can't be the judge. Answers must come from posts the team made or verified.
Not every fake is AI	Real photos with false captions and edited real photos fool people too. The content must include them.
"Fake to everything" looks smart	A student who calls everything fake scores well on fakes. Include real posts and measure skill and bias separately.
The effect fades	Research on "inoculation" games shows learning fades over weeks, so the game is a repeated practice (a gym), not a one-off lesson.
Users are minors	Minimal data, no public ranking, no fakes of real classmates, teachers or public figures.

The goal, made achievable. "Train students to spot every AI model" is impossible. The real goal: students catch cheap fakes by eye, know when they **can't** tell, and check before they believe or share.

2 · THE SOLUTION

What FakeSpotter Arena is

A web game. Each round shows posts from a realistic feed. The student decides **Real** or **Fake**, says how sure they are, and writes one sentence on **why**. If they can't tell, they press **"Can't tell, check it"** and use verification tools. Then the game reveals the answer, what gave it away, and which manipulation trick was used. Every answer feeds a teacher dashboard.

See a post
image, edited photo, text or voice

→

Decide
Real / Fake, confidence, why

→

Check (optional)
account, reverse search, reports, watermark, detector

→

Reveal
answer, clues, trick, AI coach feedback

→

Teacher dashboard
who is at risk, which tricks work

Gym, not gate

A **gate** blocks fakes before people see them (filters, detectors). A **gym** trains people to handle fakes themselves. Gates always leak; our own tests below show the best detectors miss real cases. FakeSpotter is a gym.

Two kinds of posts

Spot cheap fakes with visible flaws, plus normal real photos: trains the eye.

Check flawless fakes, edited photos, AI text, false captions, strange-but-real photos: only checking settles them.

Who uses it and how

WHO	WHAT THEY DO
Student	Plays a 5–10 minute round in class or at home, gets feedback after every post, sees a personal summary at the end.
Teacher	Runs rounds through the semester, opens the dashboard to see who is too trusting, too suspicious or overconfident, and which tricks fool the class.
Jury (pitch)	Plays a 4-post quick round live. Usually gets fooled by the flood image, sees why, sees their results land in the dashboard.

3 · REAL TEST RESULTS (11 NOV 2026)

What real detectors said about our posts

Run on the actual demo files with Hive's public AI detector, Google SynthID (inside Gemini) and GPTZero.

POST	TRUTH	DETECTOR RESULT	DETECTOR WAS
Café on Northern Avenue (AI image)	Fake	Hive 99.9% AI, source Gemini	Right
Lake Sevan "no filter" (AI image)	Fake	Hive 99.9% AI	Right
Flooded Republic Square (AI image)	Fake	Hive 99.9% AI · SynthID: "most or all" made with Google AI	Right
Opera "on fire" (real photo, AI-edited)	Fake	Hive 99.8% AI · SynthID: "part of this image" edited	Right
Cascade, Ararat cap cloud (real photos)	Real	Hive 0.4% and 0% AI	Right
Free iPhone queue (real 2001 photo, false caption)	Fake	Hive 0.1% AI: the photo is real, the lie is in the caption	Wrong
Saturday-school decree (AI-written, Armenian)	Fake	GPTZero: **100% human** in Armenian, 100% AI in English	Wrong

Takeaway for the pitch: detectors caught every AI image, but missed a lying caption and called AI-written Armenian text 100% human. SynthID survived screenshots and cropping. Tools help; trained people are still needed.

What to build: component by component

For each component: what it does, the solution, the 6.5-hour version, and what to watch out for. Components 1–6 are the core; 7–10 make it win.

01 Content bank — Core

What it does	Holds every post with its known answer. This is the answer key: the game never asks a detector who is right.
Solution	One JSON file. Each post: media, fake post context (account, caption, likes), answer, category, tier (spot/check), trick, visible flaws with x/y positions, check results with their source, explanation, credit.
Content mix	AI images (Gemini, ChatGPT, etc.), one AI-edited real photo, one AI-written text post (Armenian works best), one AI voice, a real photo with a false caption, real photos that look fake, ordinary real photos.
6.5 h version	9–16 posts. Real photos: the team's own phone photos or Wikimedia Commons (free licenses, credit the author). Fakes: one-sentence prompts in an image generator.
Watch out	No real, identifiable people in fakes. Voice clone only of a teammate who agrees. Keep original files: re-saving strips metadata.

02 Game loop — Core

What it does	Shows a post the way it appears online, collects the decision, confidence (1–5) and a one-sentence reason.
Solution	Post card (avatar, account, caption, media, likes). Real / Fake buttons, confidence scale, reason box with quick-pick chips. Progress bar. A 4-post "quick round" for demos and a full round.
6.5 h version	Single web page, any framework the team knows. Items load from the JSON file. No accounts: a nickname is enough.
Watch out	Don't show whether a post is Spot or Check before the answer, or it gives the answer away.

03 Check panel — Core

What it does	Teaches verification habits. Available on every post, costs 2 points (checking takes time in real life too).
Solution	Five tools: Who posted it? · Reverse image search (or "search the exact words" for text) · Does anyone else report it? · Watermarks & credentials (SynthID, Content Credentials) · Ask an AI detector. Each finding is labeled Real check or Scenario detail.
6.5 h version	Results are pre-written in the JSON. Technical checks are real tool outputs recorded beforehand; story context (account age, news) is invented for the fictional post.
Watch out	Some results must be inconclusive ("No Content Credentials" proves nothing), so students learn to combine clues.

04 Verification engine (the "real thing") — Core

What it does	Makes the description's claim true: real forensic tools sit behind the explanations.
Solution	Run real tools on each file and store the outputs: Hive AI detector (images; also reports Content Credentials), Google SynthID check in Gemini (images from Google tools), GPTZero (text), reverse image search (Google Lens, TinEye), Wikimedia Commons records for real photos.
Rule	**The detector is never the judge.** It's one clue. The game shows when it's wrong (false caption, Armenian text).
6.5 h version	Pre-compute everything; no live API calls during the pitch. Later: call the Hive / SynthID APIs automatically when a teacher adds a post.
Watch out	Gemini's SynthID check can hang: use a new chat per image and the fast model.

05 AI coach — Core

What it does	Grades the student's **reasoning**, not just the answer: "Right answer, but the hands look fine. The real giveaway is that no news outlet reports it."
Solution	One LLM call per answer. The prompt includes the correct answer, the key clue, the trick and the student's reason; the model returns JSON: quality (strong / partial / weak / guess), skill shown, 1–2 sentences of feedback.
6.5 h version	Any LLM API the team has. Text-only (no image sent), so it can't invent flaws. +3 bonus points for strong reasoning.
Watch out	Always have a fallback (keyword matching against the clue list) if the API is slow or down. The demo must never hang.

06 Reveal and learning — Core

What it does	Turns every answer into a short lesson.
Solution	Correct / Fooled with points · the answer and category · **trick used** (false urgency, fake authority, doctored photo, reused old photo, impersonation, engagement bait, fake authenticity) · numbered circles on the visible flaws · for edited photos a before/after slider against the original · the full check list with what the student found · image credit.
6.5 h version	Flaw circles from x/y percentages in the JSON; slider is two stacked images with a CSS clip.

07 Scoring — Supporting

Solution	Right +10 · strong reason +3 · using Check −2 · wrong while "Fairly sure" or "Certain" −10. Rewards knowing when you can tell and when you can't, and punishes overconfidence.

08 Teacher dashboard

Core

What it does	Shows who is vulnerable and how the class changes over the semester.
Solution	Signal detection theory per student: **skill** (d′, can they tell real from fake) and **bias** (c, do they lean "real" or "fake"). Profiles: Sharp · Too trusting · Too suspicious · Still learning, each with what to practice next. Flags: overconfident, right for the wrong reasons. Class views: accuracy over time for Spot vs. Check posts, posts that fooled the class, tricks that fool the class, checking habit.
Formula	`H=(hits+0.5)/(fakes+1); F=(false alarms+0.5)/(reals+1); d′=z(H)−z(F); c=−(z(H)+z(F))/2`
6.5 h version	Fictional demo class (about 24 students, 4 sessions) generated by a script and clearly labeled, plus the jury's live results on top.
Watch out	Never a public ranking. Profiles describe skills to practice, not labels like "most gullible".

09 Privacy and ethics

Supporting

Solution	Class code + nickname, no emails or accounts for students. Only the teacher sees names. No fakes of real classmates, teachers, politicians or celebrities. Consent for any voice clone. Licensed and credited real photos; edits of CC BY-SA photos shared under the same license.

10 Pitch mode

Supporting

Solution	Quick round of 4 posts (easy catch → flawless flood fake → edited Opera with before/after → real photo that looks fake), then the results screen with the "same posts through real detectors" table, then the teacher dashboard. Everything runs offline from local files.

The 6.5-hour plan

TIME	WHAT HAPPENS	MENTOR CHECK
0:00–0:30	Agree on the demo moment, scope and roles. Fix the JSON format for posts.	Is the scope small? Is anyone planning to train a model? Cut it.
0:30–2:00	In parallel: content person makes fakes and collects real photos; developers build the game screen with 3 placeholder posts; one person tests the LLM coach on its own.	Do the fakes fool *you*? Do real photos have sources?
2:00	Checkpoint 1 Click through the whole game end to end, even if ugly.	If not, cut the check panel and audio now.
2:00–4:00	Real posts in; check results recorded with real tools; reveal screen; dashboard with demo data; coach wired with fallback.	Explanations specific, not "looks AI"? Real vs. scenario labels honest?
4:00	Checkpoint 2 Full loop: play → reveal with coach → dashboard.	If yes, allow one extra (before/after slider, trick panel, voice).
4:00–5:15	Test posts on other participants, replace ones everyone gets right; polish; reset-demo button.	Watch for scope creep.
5:15	Feature freeze No new code.	Be strict.
5:15–6:30	Pitch script, 3+ rehearsals, backup video, test on the presentation laptop.	Does the demo run in under 2 minutes, offline?

Roles (team of 3–5)

- **Content:** fakes, real photos, clues, check results, real tool runs. Full-time job.
- **Frontend (1–2):** game, reveal, dashboard.
- **AI / backend:** coach prompt and fallback, answer storage.
- **Pitch:** from hour 3, script, demo flow, backup video.

Red flags

- Training a detector from scratch.
- Only AI content, no real or miscaptioned posts.
- The detector decides right and wrong.
- Fakes of real people; public student ranking.
- Demo depends on Wi-Fi and live APIs.
- Pretty UI, no working loop at hour 2.

Prototype: FakeSpotter Arena demo (Claude artifact, private until shared). Images: Google Gemini (AI), Wikimedia Commons (Marcin Konsek, Cetintas555, Blake Patterson, Matthias Süßen; CC licenses). Detector results recorded 11 November 2026.

The 3-minute pitch

TIME	WHAT HAPPENS
0:00	"Before we explain anything, play." A judge plays the quick round on the laptop.
0:20	The judge catches the café fake (garbled menu) and feels confident.
0:40	The flood image: most judges say Real and get fooled. Show it took one sentence and 20 seconds to make.
1:10	The Opera: Check panel shows SynthID and the before/after slider against the original photo.
1:40	Results screen: "the same posts through real detectors": right on every AI image, wrong on the lying caption and the Armenian AI text.
2:10	Teacher dashboard with the judge's results among the class: profiles, tricks, progress.
2:40	Close: what's next and why it lasts.

> **"The best detector we tested caught every AI image, but called our AI-written Armenian post 100% human, and can't read a lying caption. That's why we built a gym, not a gate."**

Questions to prepare for

QUESTION	ONE-SENTENCE ANSWER
Doesn't this just make kids distrust everything?	No: real posts are in every round, and the dashboard measures bias separately from skill, so "too suspicious" is flagged too.
Is the detection engine real?	Yes: every technical result comes from Hive, SynthID or GPTZero run on these files, labeled as such. But it's a clue, not the judge.
How do you keep up with new generators?	We train habits that work on any model, track tricks (which change slowly), and add new posts monthly, including ones teachers and students bring in.
How do you know it works?	Inoculation-game research (Bad News, Go Viral) plus our own before/after accuracy; next step: test on generators students never practiced on.
Student privacy?	Class code and nickname, no student accounts, teacher-only names, no public ranking.
Why not just use a detector?	Our own test: detectors missed the false caption and the Armenian AI text. People who check are still the last line.

After the hackathon (roadmap)

- **Live classroom mode:** the whole class votes on phones, the room sees the split.
- **"Make a fake" mode:** students create misleading posts and classmates try to catch them (the strongest learning in the research).
- **Weekly booster rounds** with variants of posts each student got wrong.
- **Unseen-generator test** to prove skills transfer to new models.
- **Teacher content editor** with automatic Hive / SynthID checks on upload.
- **Armenian interface** and more Armenian- and Russian-language posts.